Received: 3 September 2021 | Revised: 3 December 2021 | Accepted: 26 December 2021

IET Generation, Transmission & Distribution

DOI: 10.1049/gtd2.12392

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH PAPER

Flexibility provision of residential energy hubs with demand response applications

Sara Ostovar | Moein Moeini-Aghtaie | Mohammad Behzad Hadi

Department of Energy Engineering, Sharif University of Technology, Tehran, Iran

Correspondence
Moein Moeini-Aghtaie, Department of Energy Engineering, Sharif University of Technology, Tehran 11365-11155, Iran.
Email: moeini@sharif.edu

Abstract

Smart grids help local distribution companies (LDCs) facilitate the communication between grid operators and residential prosumers. While the prosumers are striving to optimize their daily load profile by home energy management systems (HEMSs) deployment, the LDCs attempt to enhance the operation of the grid in an efficient way and decrease network losses. This paper attempts to develop a new bi-level probabilistic optimization framework wherein an HEMS optimizes its respective daily load profile, and determines its flexibility provision, which is communicated to an LDC. In the proposed framework, a decentralized approach is used to achieve the flexibility product, which is the modulation of energy, through the incentive-based demand response (DR) programs. Finally, the LDC applies the prosumer's flexibility' offers to optimize its operational performance. The two-point estimate method (2PEM) is employed to model the uncertainties. The applicability of the framework is demonstrated by applying it to a system with one residential feeder.

1 | INTRODUCTION

The power systems' net-load variability and uncertainty would aggravate significantly as the penetration of renewable energy resources (RERs) in power systems increases [1–3]. In response, power system researchers have introduced a new concept, namely flexibility, to quantitatively and systematically deal with unavoidable features of RERs. The term flexibility describes the ability of a power system to cope with variability and uncertainty in both generation and demand while maintaining a satisfactory level of reliability at a reasonable cost over different time horizons [4]. Recently, flexibility has been involved in operational and planning studies of various sectors in power systems. However, the presence of RERs in these systems may also overshadow the flexibility of power systems. Variability in output generation of Distributed Generations (DGs) may cause some problems for distribution system operators (DSOs) for example, reverse power flows, congestions, voltages droop, and network losses which need to be covered by flexibility services for example, DSO congestion management, and DSO power quality control [5–7].

Recently, the emergence of residential prosumers with great DR potentials has created new opportunities for DSOs in providing the system with different services of flexibility in distribution systems [8]. These DR potentials (e.g., available positive and negative power capacities, and available energy storage capacity in case of shiftable loads) are subjected to limitations due to the controllability of loads and behaviours of prosumers. These limitations have a significant impact on the result of the implementation of DR applications, that is, energy-saving and reduction, which significantly affect the system's flexibility level. It is shown that using an HEMS leads to the well-organized contribution of residential prosumers in DR applications to provide flexibility [9, 10]. The HEMSs calculate and send prosumers' load profile signals to an LDC. So that, LDC realizes each prosumer's flexibility by coordinating the controllable loads. Coordinated energy management in the smart grid is studied in either centralized or decentralized manners. With centralized coordination, decision-making is performed by one superior entity for example, aggregators, LDCs. Centralized coordination proposes efficient costs and effective results, however, it requires detailed information about residential energy hubs, which is not practical [11] and also tends to cause a heavy computation burden [12]. Moreover, prosumers' privacy security concerns may arise as well, which is solvable by direct control of the appliances by themselves [13–15]. On the other hand, with decentralized coordination, end-use prosumers are independent decision-makers who control their own electricity profile under the influence of a central entity. Decentralized approaches decomposing the problem into subproblems improve the scalability and the privacy issues [16]. In [17], the existing decentralized approaches for DR programs are classified. This classification is based on the mathematical complexity of the models for appliances. The authors in [18, 19] present decentralized approaches to optimize the power consumption of residential loads using individual HEMSs. Although the proposed approaches have been shown effective in load shaping and energy cost reduction, they face the long-time delay requirement for finding a solution, mainly due to sequential prosumers' decision making. Besides, in the previously mentioned studies, the associated optimization problem is formulated as non-linear programming, which may reduce the possibility of obtaining the global optimum. Furthermore, it is assumed in these works that the objective of prosumers is always dependent on the objective of the central entity, for example, aggregators, LDCs, which is not always the case.

In this paper, a bi-level probabilistic optimization framework is proposed to coordinate residential prosumer contribution in providing flexibility services to an LDC to deal with the abovementioned issues. To reach this goal, at first, a residential prosumer's building is mathematically modelled as a Residential Energy Hub (REH) considering different electrical and heating appliances, CHP units, PHEVs, RDERs, and ESDs. To reach the optimal operating strategies for the REH, an HEMS is introduced. In addition to the operating strategies of REHs, the HEMS determines interactions with the LDC, that is, power to be sold to or purchased from the LDC. In the proposed framework, to provide flexibility services, the HEMS tries to approximate the prosumer's load profile to the LDC's desired load profile without loss of the prosumer's desired social welfare level. The aforementioned problem is modelled as a bi-level optimization model. Afterward, a new decentralized coordination approach is introduced, which optimizes participation of flexibility services through incentive-based DR programs. The proposed coordination framework in this paper extends the existing models from several directions. First of all, the LDC and prosumers have separate objectives. The LDC intends to modify system load profile and prosumers would like to maximize their own social welfare levels. Moreover, in the proposed model, prosumers solve the load reschedule problems non-sequentially. Therefore, exhaustive time delays are avoided. Besides, an equivalent linear single-level model of a bi-level nonlinear coordination framework is obtained that guarantees the global optimum and requires less computational time and a lower degree of complexity in the calculation. In addition, using the proposed framework, different end-use prosumers can consider their preferences in the controllability status of their appliances. The provided flexibility according to the proposed framework could potentially enhance the operations of grids by improvements in bus voltages, flattening the load profile of the system, reduction in losses, and the reduction in LDC's costs of operating. Accordingly, in summary, the contribution of this paper, compared with previous studies, can be summarized as follows:

- A new coordination framework for residential prosumers' contribution in providing flexibility services is developed in which the LDC and prosumers have separate objectives. The LDC intends to modify the system load profile, and prosumers would like to maximize their social welfare levels. In the literature reviewed, it is assumed that the objective of prosumers is always dependent on the objective of the central entity, for example, aggregators, LDCs, which is not always the case;
- In the proposed model, prosumers solve the load reschedules problems non-sequentially. This removes the barriers of exhaustive time delays. In the previously mentioned studies, they face long time response delays for finding a solution, mainly due to sequential prosumers' decision making;
- An equivalent linear single-level model of a bi-level nonlinear framework is presented that guarantees the global optimum and requires less computational time and a lower degree of complexity in the calculation. In the reviewed studies the associated optimization problem is formulated as nonlinear programming, which may reduce the possibility of obtaining the global optimum;
- Different end-use prosumers can consider their preferences in the controllability status of their appliances in the proposed framework which can be easily directly applied to practical cases to satisfy the prosumers' objectives.

The rest of the paper is structured along these lines. The structure and formulation of the proposed framework between the HEMS and the LDC have been brought in Section 2. Section 3 covers simulation results and a brief discussion about them. Finally, Section 4 concludes this paper.

2 | METHODOLOGY

The extreme installation cost of the smart homes' devices will charge the households with high prices. Therefore, it is necessary for an HEMS to have participation in flexibility services to financially be attractive for the related prosumer and LDC. As depicted in Figure 1, an HEMS is a residential controller which schedules a prosumer with all its respected appliances, ESDs, RDERs, and the power interchanges with the external grid, considering end-users' objectives and preferences. The HEMSs' objective function is the energy cost minimization, energy consumption minimization, or the related prosumer's social welfare maximization. The integration of an HEMS within a prosumer makes it a flexibility service provider and therefore its financial viability is increased [20].

2.1 | Overall procedure

In this work, a decentralized approach for the participation of prosumers in providing flexibility services is introduced. The overall procedure is illustrated in Figure 2. The proposed procedure is consisting of two main levels. At the first level, the LDC declares electricity and gas prices to the prosumer, that

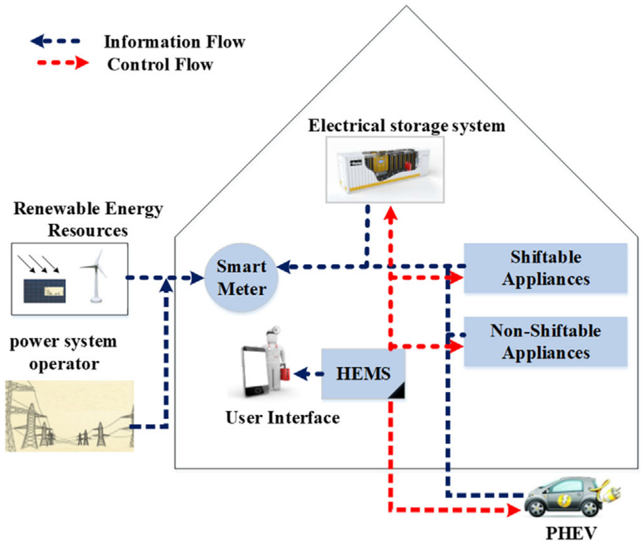

FIGURE 1 A typical illustration of HEMSs architecture

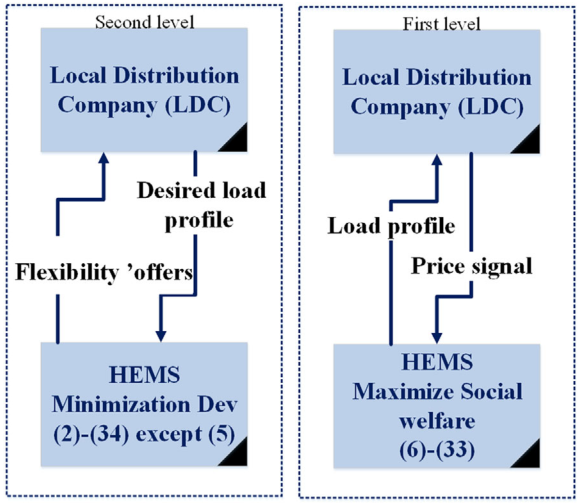

FIGURE 3 The decentralized method of the first and second levels

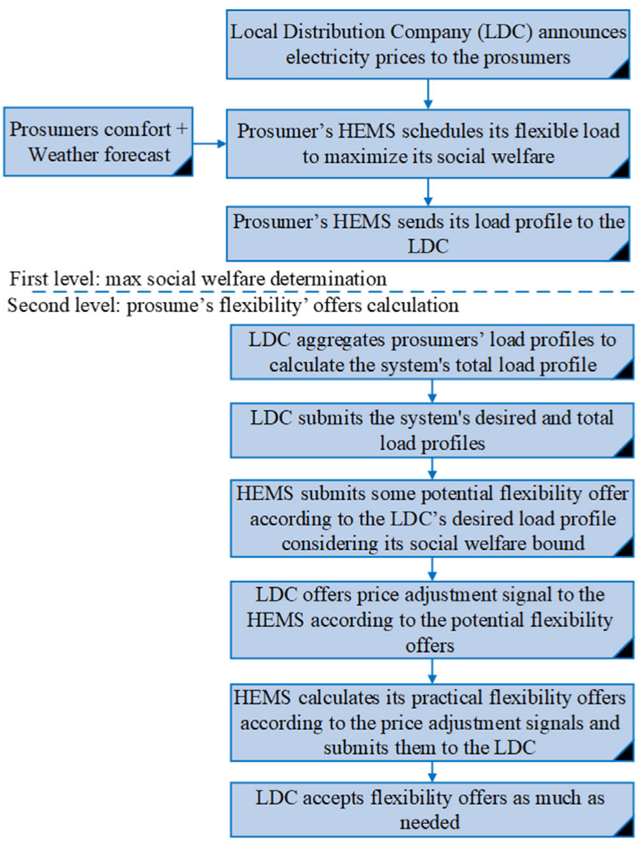

FIGURE 2 The two-level decentralized approach

is, the HEMS module. Then, the HEMS maximizes the prosumer's social welfare and sends its optimized load profile to the LDC. The obtained load profile of the prosumer is considered as "baseline demand". The first level's decentralized procedure is illustrated in Figure 3. The HEMS module records its schedules and social welfare amount at the end of the first level. Then, the LDC records the received load profile. The recorded data are then imported to the second level as input data.

The second level (which is shown in Figure 3) of the algorithm aims to achieve suitable potential flexibility' offer based on the LDC's objectives without compromising prosumers' costs and social welfare. In the context of the present work, the potential flexibility (in kWh) of REHs is defined as the possible load variation amount by a prosumer at a definite time interval. In the second level, the LDC calculates the system's total load profile, then announces its desired load profile to the HEMS. By receiving the LDC's desired load profile signal, the prosumer tries to bring its load profile closer to the LDC's. The obtained load profile is defined as the "desired load profile". The potential flexibility available from an REH, at a given time interval, is calculated through the difference between the "desired load profile" and the "baseline demand" of the prosumer. Using the price adjustment programs, the HEMS investigates whether it can propose new potential flexibility offers. It calculates practical flexibility proposals to the LDC through the difference between the two values of hourly potential proposals. The practical flexibility (in kWh) is defined as the load variation amount of any prosumer that is willing to be made available to the LDC.

The monetary incentives provided by the incentive-based price adjustment programs as DR programs are the primary motivation for prosumers to propose flexibility' offers. At the end of the second level, prosumers' load schedules dedicated to both the LDC's and the prosumer's objectives are obtained. The prosumer's social welfare is maximized then, and the system's total load profile is improved. The negative and positive flexibility of prosumers indicates that a possible load increase and reduction, respectively. The prosumer's flexibility can significantly improve the load profile of the system, bus voltages, and network losses [20]. According to the proposed decentralized approach, the load management problem is locally solved, therefore, releasing the information of the prosumers' appliances and preferences is not needed. Hence the prosumers' privacies are successfully protected.

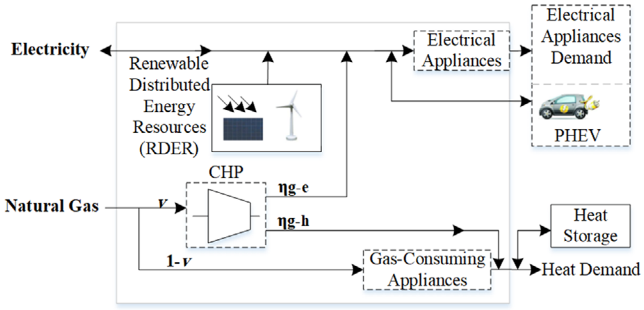

FIGURE 4 The overview of the proposed REH

2.2 | Bi-level model

The bi-level model of the problem is established to calculate the prosumer's flexibility' offers and provides flexibility services to an LDC. The proposed model includes an upper-level problem and a lower-level problem. The upper-level optimization problem modifies the system's total load profile while the lower-level tries to maximize the individual prosumers' social welfare. The proposed model formulation is formulated as a bi-level model because the total system's load profile could be reshaped only by taking the prosumers' preferences and objectives into account.

2.2.1 | Proposed renewable-based REH

In this work, a residential prosumer is deemed as an REH which is depicted in Figure 4. In this illustration, the electrical demand is supplied either by the grid, the CHP unit's output power, and RDERs. When it is needed, the PHEV could return the stored electrical energy, for as much as its vehicle to grid (V2G) capability, to the REH to supply electrical appliances. The power flow between the grid and the REH is bidirectional and it demonstrates that the unconsumed electricity in the REH which is not either stored, could be sold to the grid. In the REH understudy, the received natural gas as input of the REH is divided into two ways, one way to the input of the CHP unit, and another to gas-consuming appliances. Moreover, the gas-consuming appliances, the CHP unit's output heat, and the Heat Storage Devices (HSDs) are used to supply the heat demand.

2.2.2 | Bi-level model formulation

In this work, the flexibility index is achieved by approximating the end-user load profile to the desired load profile of the LDC, which is considered as the objective function of the upper-level problem as follows.

$$\min\ Dev = \sum_t |E_{grid}(t) - E_{des}(t)| \tag{1}$$

$E_{grid}(t)$ and $E_{des}(t)$ are the daily and the desired load profiles from the LDC's viewpoint at time t, respectively. According to (1), the objective function is defined as the summation of all deviations of the load profile of the system from the LDC's desired profile which needs to be minimized. Since the objective function is non-linear, the linearization procedure is described as follows.

$$E^{\text{pos.}}(t) - E^{\text{neg.}}(t) = E_{grid}(t) - E_{des}(t) \tag{2}$$

$$0 \le E^{\text{pos.}}(t) \le y_t E_{dev}^{up}, \quad \forall t \tag{3}$$

$$0 \le E^{\text{neg.}}(t) \le (1 - y_t) E_{dev}^{up}, \ \forall t \tag{4}$$

$E^{pos.}$ and $E^{neg.}$ are values dedicated to the excess deviation of load from the desired value and shortage of the load from the desired load, respectively. y_t is a binary variable and E_{dev}^{up} is the upper limit for load deviation from the desired load. The prosumer's social welfare is taken into account using the following sub-problem:

$$\text{Max}\ sw = \text{Max}(Utility - \text{Cost}) \tag{5}$$

It is assumed that the utility function does not decrease, implying that users could complete more important tasks (e.g., to keep the temperature in a predetermined range) if they consume more power. Hence, as the energy price increases, lesser important tasks are ignored by the user. Moreover, the user switches to lower energy consumption operation modes as the energy price increases. This issue indicates that the user's energy valuation is regulated through parameter w, as w increases, the utility function is increased [21]. According to the above discussion, the utility function is considered as follows.

$$U(E_{app}, w) = \begin{Bmatrix} wE_{app}, \text{if}\ 0 \le E_{app} \le \frac{w}{\alpha} \\ \frac{w^2}{2\alpha} \quad , \text{if}\ E_{app} \ge \frac{w}{\alpha} \end{Bmatrix} \tag{6}$$

α and E_{app} are predetermined parameters by the end-user and the amount of electrical energy consumed by the end-user, respectively. The same procedure can be employed for gas-based utilities. The end-user is assumed to be a price taker entity and its dedicated cost function is considered as (7).

$$C = \sum_{t=1}^{24} (\lambda_{elec}(t) \times E_{grid}(t)) + (\lambda_{gas}(t) \times Gas(t)) + \sum_t (VOLL \times L_{shed}(t)), \tag{7}$$

λ_{elec} and λ_{gas} represent respectively the price of electricity and natural gas. In this paper, it is assumed that the electricity prices are known and cannot be changed once they are announced. Therefore, they are among the input parameters of the model. *Gas(t)* stands for the natural gas demand of the prosumer at time *t*. L_{shed} and $VOLL$ are the amounts of load shedding and the Value of Lost Load, respectively. C is the energy expenditure of the prosumer. The prosumer's HEMS minimizes the total energy expenses. This value is related to price selection schemes that is, flat, time-of-use, and real-time tariffs. The HEMS optimization problem constraints related to the equations of the electricity and heat flow in the proposed REH are as follows.

$$E_{grid}(t) + \eta_{DC/AC} E_{RDER}(t) + E^{\mathrm{CHP}}(t) + \eta_{dch}^{PHEV} E_{dch}^{PHEV}(t) = \left(\frac{1}{\eta_{eapp}}\right) E_{app}(t) + \left(\frac{1}{\eta_{ch}^{PHEV}}\right) E_{ch}^{PHEV}(t) - L_{shed}(t),\ \forall t \tag{8}$$

$$H_{\mathrm{appli.}}(t) + H^{\mathrm{CHP}}(t) + \eta_{dch}^{s} H_{dch}^{s}(t) = H_d(t) + \left(\frac{1}{\eta_{ch}^{s}}\right) H_{ch}^{s}(t), \forall t \tag{9}$$

$$H_{\mathrm{appli.}}(t) = (1 - v(t)) Gas(t) \eta_{\mathrm{gas}}^{\mathrm{app}} \tag{10}$$

$$E^{\mathrm{CHP}}(t) = v(t) Gas(t) \eta_{\mathrm{gas-elec.}} \tag{11}$$

$$H^{\mathrm{CHP}}(t) = v(t) Gas(t) \eta_{\mathrm{gas-heat}} \tag{12}$$

Equations (8) and (9) represent the power and heat balances. $E_{grid}(t)$ is a negative value if the electricity is returned to the grid. E^{RDER} is the RDER's provided electrical energy. E_{ch}^{PHEV}, E_{dch}^{PHEV}, H_{ch}^{S}, and H_{dch}^{S} are the charged and discharge energy of the PHEV battery and the energy stored in and supplied by the heat storage unit, respectively. $H_{appli.}$ and H_d are respectively, the gas-consuming appliances' heat generation and total household heat demand. Equation (10) determines the gas-consuming appliances' output heat which is calculated by the received natural gas amount at each hour of a day. In (11) and (12), $E^{CHP}(t)$ and $H^{CHP}(t)$ are calculated by $\nu(t)$, $\eta_{gas-heat}$, and $\eta_{gas-elec.}$ which are the dispatch factor, gas to heat, and gas to electricity conversion efficiencies of the CHP unit, respectively. It is noticeable that to linearize the product of two decision variables, the binary expansion technique is applied.

At each time step of operation studies, the prosumer's electricity and heat demand are defined as the summation of controllable $((E^{cont.})$ or $(H^{cont.}))$, and uncontrollable $((E^{uncont.})$ or $(H^{uncont.}))$ appliances' consumptions which are presented as follows.

$$E_{app}(t) = E^{cont.}(t) + E^{uncont.}(t) \tag{13}$$

$$H_{appli.}(t) = H^{cont.}(t) + H^{uncont.}(t) \tag{14}$$

The consumption level of the uncontrollable appliances is predefined values. The water heater, and electrical heating/cooling systems energy consumption level is controlled in operational intervals. The controllable electrical appliances operational constraints, according to the [22, 23], are as follows.

$$E_k^{\min}(\mathrm{t}) \leq E_k(t) \leq E_k^{\max}(t), \forall \mathrm{k} \in \mathrm{K}, \forall t \in AI_k, \tag{15}$$

$$\sum_{t \in AI_k} E_k(t) \geq E_k, \forall \mathrm{k} \in \mathrm{K}, \tag{16}$$

where E_k is the electrical energy consumption of appliance k. $E_k^{min}(t)$ and $E_k^{max}(t)$ are the allowable minimum and maximum energy consumption of appliance k at hour t, respectively. These values are extracted from the controllable appliances' historical energy consumption data. These equations are applicable to gas-consuming appliances similarly. In (15), at every definite time step, the energy consumption of each electrical appliance is limited by the maximum and minimum values. In line with [22, 23], the prosumer's comfort is considered as the range of each controllable appliance's energy consumption in (15). It means that the $E_k^{min}(t)$ and $E_k^{max}(t)$ are considered according to the appropriate temperature range which is achieved at each hour for prosumers. Equation (16) identifies the appliances' total electrical energy consumption is greater than a certain value, that is, E_k to keep the prosumer's desired comfort level. E_k is determined in line with the prosumer's desired temperature. The controllable appliances' total energy consumption at hour t is calculated as follows.

$$\sum_{k \in K} E_k(t) = E^{\mathrm{cont.}}(t), \forall \mathrm{t} \tag{17}$$

Prosumer's comfort can be modelled using [24, 25] taking the following actions. First, the prosumer has to decide which appliances are controllable. Second, the prosumer defines the controllable appliances' operational constraints.

During available times of the PHEV, it acts as an electrical storage device (ESD) [26]. The PHEV battery charging and discharging rates are limited with certain values which are shown by the following constraints:

$$E_{ch}^{PHEV}(t) \leq E_{ch}^{\max} \tag{18}$$

$$E_{dch}^{PHEV}(t) \leq E_{dch}^{\max} \tag{19}$$

E_{ch}^{max} and E_{dch}^{max} are the maximum charging and discharging rate of the PHEV at each hour, respectively. It is assumed that the PHEV is on travel at certain hours of the day. d is the departure hour of PHEV and c is the arrival time. It is considered that charging and discharging have not occurred in the time interval of $[d,c]$. The PHEV battery charging level is determined before and after the travel time, which is:

$$E^{PHEV}(t) = \begin{Bmatrix} E_0^{PHEV} + \sum_{h=1}^{t} (E_{ch}^{PHEV}(h) - E_{dch}^{PHEV}(h)) \\ , \forall \mathrm{t} \leq \mathrm{g} - 1 \\ E_0^{PHEV} + \sum_{h=1}^{t} (E_{ch}^{PHEV}(h) - E_{dch}^{PHEV}(h)) \\ - E_{out}^{PHEV} \quad , \forall \mathrm{t} \geq c + 1 \end{Bmatrix}, \tag{20}$$

where E_0^{PHEV} and E_{out}^{PHEV} are the battery of the PHEV charge level at the initial hour of the day and the PHEV electrical energy consumption during the travel time, respectively. Moreover, h is counter for t, and $E^{PHEV}(t)$ is the charge level of the PHEV battery at the end of each hour. The PHEV battery charge level is capped with its capacity ($capacity^{PHEV}$) which is modelled by the following constraint:

$$E^{PHEV}(t) \leq \text{capacity}^{PHEV} \tag{21}$$

The fully charged PHEV battery is expected by the prosumer before the time of departure ((d-1)th hour) which is modelled as follows.

$$E^{PHEV}(d-1) = \text{capacity}^{PHEV} \tag{22}$$

The battery charge of the PHEV at the end of the day should not be less than the initial amount which is determined by the following constraint:

$$E^{PHEV}(24) \geq E_0^{PHEV} \tag{23}$$

The constraints associated with the HSD are presented as follows. The heat energy that is stored in the HSD, $H^s(t)$, can be considered at the end of each hour by the following constraint:

$$H^s(t) = H_0^s + \sum_{h=1}^{t} H_{ch}^s(h) - H_{dch}^s(h), \tag{24}$$

where H_0^s is the HSD's heat energy level at the initial hour of the day. The HSD charge and discharge rates are limited by the predefined values, which are modelled by the following constraints.

$$H_{ch}^s(t) \leq H_{ch}^{\max} \tag{25}$$

$$H_{dch}^s(t) \leq H_{dch}^{\max} \tag{26}$$

H_{ch}^{max} and H_{dch}^{max} are the maximum HSD's charging and discharging rates at each hour of the day. Moreover, the HSD remained heat energy at the end of the day (24th hour) should not be less than the initial heat level, which is modelled as follows.

$$H^s(24) \geq H_0^s \tag{27}$$

The CHP's natural gas input is limited with the $E_{max}^{CHP}(t)$, which are modelled by the following constraint.

$$v(t)Gas(t) \leq E_{\max}^{CHP} \tag{28}$$

The involuntary load shedding constraint is as follows.

$$0 \leq L_{shed}(t) \leq \left(\frac{1}{\eta_{eapp}}\right)E_{app}(t) + \left(\frac{1}{\eta_{ch}^{PHEV}}\right)E_{ch}^{PHEV}(t) \quad ,\forall t \tag{29}$$

2.2.3 | Solar panels output uncertainty model

Several approaches have been presented in the literature to model the uncertainties associated with the output generation of renewable-based generating units [27–43]. However, some works have taken a deterministic approach when dealing with the output power of solar panels [27–29]. Moreover, in [30, 31] robust optimization method is used to consider the uncertainties, therefore the worst scenarios are characterized in these works. On the other hand, distributional robust optimization (DRO) are used in the literature which is a modelling approach that assumes only partial distributional information [32], whereas probabilistic optimization assumes complete distributional information. As the probability distribution of the solar radiation uncertainty is known, a probabilistic approach is preferred in this work. In some cases, when a probabilistic approach has been taken, a high computational burden is involved [33]. Among them, Monte Carlo simulation (MCS) is the most popular one. However, the MCS approach usually needs high computational efforts to reach highly accurate results [34]. In order to address this limitation, some studies have used approximate methods to model uncertainty for example, the truncated Taylor series expansion method [35]; the discretization method [36]; the common uncertain source method [37]; the first-order second-moment method (FOSMM) [38]; the cumulant method [39]; and the point estimate method (PEM) [40, 41]. The main idea behind these methods is to use approximate formulas for calculating the statistical moments of a random quantity that is a function of random variables, as opposed to a more accurate but computationally demanding MCS approach. The PEM has been introduced as an approximate method with an acceptable level of accuracy and efficiency to calculate different moments of random variables [42, 43]. The 2PEM as a variant of PEM is employed in this paper to deal with the uncertainties associated with the output power of solar panels. One of the main advantages of this probabilistic modelling procedure of the uncertainties is reducing the computational burden. It is assumed that $E_{solar}(t)$ is the hourly solar panels' output power and the vector X as the input random variables' vector is considered by the following constraint.

$$X = [E_{\text{solar}}(t=1), E_{\text{solar}}(t=2), \ldots, E_{\text{solar}}(t=T)] \tag{30}$$

The solar generation random parameter is replaced by concentration points at each hour in the 2PEM algorithm. The concentration points at each hour are two deterministic points that are placed on the two sides of the corresponding solar radiation mean value. The concentration points of each solar generation random parameter are symbolized by $E^1_{solar}(t)$ and $E^2_{solar}(t)$. Therefore, 48 scenarios are needed to evaluate in one day. In the optimization problem of each scenario, the related output power concentration point, that is, $E^k_{solar}(t)$ and solar output power mean value of other time steps, that is, $\mu E_{\ solar(t)}$ are considered, which is presented by the following constraint.

$$X = [\mu_{E_{\text{solar}(1)}}, \ldots, E^k_{\text{solar}}(t), \mu_{E_{\text{solar}(t+1)}}, \ldots, \mu_{E_{\text{solar}(24)}}] \tag{31}$$

By solving (1) subjected to (2)–(29), the optimal schedules of the end-user demand and the optimal operation of the REH are obtained. Therefore, the output variables' vector, that is, $g_{t,k}$ is determined as follows.

$$g_{t,k} = g(\mu_{E_{\text{solar}(1)}}, \ldots, E^k_{\text{solar}}(t), \mu_{E_{\text{solar}(t+1)}}, \ldots, \mu_{E_{\text{solar}(24)}}) \tag{32}$$

It is assumed that the concentration points' weights are the same. Therefore, each decision variable amount is calculated by the average of the obtained results in all of the scenarios, as follows.

$$E[g] = \sum_{k=1}^{2} \sum_{t=1}^{24} \left(\frac{1}{48} \times (g_{t,k}) \right) \tag{33}$$

2.3 | Equivalent single-level model

In this section, the bi-level model has become a single-level model. Solving the single-level equivalent problem is much easier than solving the bi-level problem. In the proposed bi-level model, the upper-level problem's optimal solution has no effect on the objective function of the lower-level problem. As a result, the lower-level problem can be independently solved. The lower-level problem's optimal solution can be used to convert the objective function (5) to a new constraint. The optimization problem is then solved in two hierarchical stages. At first, the social welfare of the individual prosumer is maximized subjected to (6)–(33). Then, employing the calculated optimum value of social welfare, (5) is replaced by the following constraint:

$$SW \geq SW^* \tag{34}$$

where SW^* is the maximum social welfare of prosumer. Constraint (34) guarantees that the prosumer's social welfare remains at its maximum values and the proposed procedure incurs no additional costs for the prosumer. The bi-level problem now becomes a single-level equivalent problem with objective function (1) subjected to (2)–(34) except (5).

3 | SIMULATION AND RESULTS

The proposed bi-level probabilistic optimization framework is examined by three different cases in this section. The following cases are conducted as:

TABLE 1 A sample maximum and minimum allowable energy consumption (kWh) for two controllable electrical appliances

Hours	E_{1Max}	E_{1Min}	E_{2Max}	E_{2Min}
1–24	2	0	2.5	1

TABLE 2 A sample maximum and minimum allowable energy consumption (kWh) for two controllable gas consuming appliances

Hours	H_{1Max}	H_{1Min}	H_{2Max}	H_{2Min}
8–17	1	0	1	0.5
18–7	5	0	5	2

TABLE 3 PHEV characteristics

$E_{ch}{}^{Max}$	$E_{dch}{}^{Min}$	$E_0{}^{PHEV}$	Cap^{PHEV}	η_{ch}	η_{dch}
1.4 (kWh)	1.4 (kWh)	3.9 (kWh)	7.8 (kWh)	0.88	0.88

First case—The bi-level framework is applied in the proposed REH model.

Second case—The first price adjustment program is added to the first case.

Third case—The second price adjustment program is added to the first case.

The proposed REH has a PHEV, an HSD, a CHP unit, solar panels, and 4 controllable electrical and heat appliances. The controllable appliances consist of an electrical water heater, an electric air conditioner system, and two gas-consuming space heaters. These controllable appliances' allowable energy consumption level is presented in Tables 1 and 2. The reported data is based on the energy consumption profile of each controllable appliance, the thermodynamic nature, and the forecasted outdoor temperature in former days [44, 45]. The PHEV is assumed to be unplugged between the 16th and 20th hours of the day. In Tables 3 and 4, the PHEV battery and CHP unit characteristics can be found, respectively [46, 47]. Moreover, the input natural gas is measured by its kWh energy content. In line with [27, 48], the heat and electrical demands of uncontrollable appliances are considered deterministic parameters. A two-level TOU tariff is used for electricity in this work. The proposed tariff is assumed to have high-level and low-level tariffs. The prices of electricity levels are considered as 14.5 ¢/kWh for the high-level tariff and 10.8 ¢/kWh for the low-level tariff. The highly-priced periods are the 7th-8th and the 19th-21th hours of the day and the other hours are low-priced periods. It is possible in all scenarios to sell the electricity to the grid at the same price. Moreover, the price of natural gas is considered 1.3 ¢/kWh in

TABLE 4 CHP unit data

η_{g-h}	η_{g-e}	$E_{Max}{}^{CHP}$
0.4	0.3	15(kw)

all operation times. The HSD capacity is assumed to be 10 kWh and the initial unit's SOC is 0.5. The capacity of the solar panels is considered to be 6 kW which is added to the REH. The efficiency of the solar panels' AC/DC converter is considered to be 0.95. The proposed framework is probabilistically run using the 2PEM method. The historical radiation data of 2019 to 2020 is used and the statistical information which is required for solar panels' output power calculation is extracted [49]. Finally, this information is fed into the 2PEM algorithm as input data.

In order to calculate the potential flexibility of the prosumer, the proposed first case is implemented in this section. At first, the HEMS schedules the baseload profile of the end-user by solving the lower-level sub-problem so that the social welfare associated with the end-user reaches its maximum value. As a result of solving the lower level problem, the amount of the dispatch factor and input natural gas of the energy hub and the charging state of the PHEV and HSD units are shown in Figure 5. As can be seen, during off-peak hours that the price of electricity is low, the HEMS buys more electricity from the grid and also charges the PHEV during these hours. Moreover, during these hours, the HSD unit is discharged to meet the heat demand and the amount of dispatch factor is reduced. During peak hours, it is preferable to sell electricity to the grid, and the amount of purchased natural gas from the grid is higher to meet electricity demand. Moreover, the PHEV battery is discharging and the HSD is charged during these hours. When there is electricity generation through the RDER unit, in addition to meeting the electrical demand, the PHEV is also charged.

Thereafter, to calculate the potential flexibility of the prosumer, it is needed to solve the upper-level problem to obtain the prosumer's desired load profile. As a result of solving the upper-level problem, the amount of the dispatch factor and input natural gas of the energy hub and the charging state of the PHEV and HSD units are also shown in Figure 5. In this case, while the social welfare of the end-user is satisfied, the HEMS tries to approach the LDC desired load profile. To compare the obtained results of the upper-level problem with the lower-level problem results, as shown in Figure 6, the HEMS is preferred to buy more electrical energy to meet the electrical demand during off-peak hours. The amounts of potential flexibility' offers in the first case are also shown in Figure 6. As it can be seen, there are no potential flexibility' offers during peak hours. This is due to the fact that it is not profitable for the HEMS to provide flexibility and lose the amount of profit it can make by selling electricity to the grid during these hours. In the 18th hour of the day, a relatively high amount of negative potential flexibility is provided, and it is worthwhile not only to reduce the amount of electrical energy purchased but also to sell its extra amount to the grid. The discharging PHEV, load shifting, and the CHP unit are used to meet the electrical demand in this specific hour. The amount of provided positive flexibility during some off-peak hours is significant. This issue is appeared because of the high differences between the LDC's desired load profile and the HEMS' baseload profile. Therefore, by solving the upper-level problem, the HEMS tries to reduce these differences by bringing its desired load profile closer to the LDC's, so more positive flexibility has been provided. The HEMS tries to meet the electrical demand through the PHEV charging and reducing natural gas input in this situation. It is worth saying that the obtained results of the potential flexibility of the residential prosumer have a similar trend with those in [20].

TABLE 5 The results of the runtime of the MCS and 2PEM approaches

	The 2PEM approach	The MCS approach
Run time	21.9 (min)	33.4 (min)

The required computational effort of uncertainty consideration using the 2PEM method affects the runtime of the problem. In order to show the tractability of the proposed model and its validation, the proposed optimization problem of obtaining the potential flexibility is run with the MSC approach for 48 most probable scenarios and the results are compared with those for the 2PEM approach which are as follows:

As it can be seen by Table 5, the problem is solved with two uncertainty modelling approaches that is, the MCS and 2PEM approaches. The global optimum solution is obtained using both approaches but the run time in the 2PEM method is became significantly improved compared to the MCS approach. The result shows the efficiency of the 2PEM method as an appropriate method for uncertainty modelling with less computational effort requirements in this paper. According to the presented results, the performance of the proposed model is validated.

Finally, by defining two scenarios, the practical flexibility proposals are calculated. These scenarios are communicated to the HEMS through the LDC in order to create incentives. By receiving price adjustment signal, the HEMS performs the bi-level optimization approach again, and the difference between the daily load profile results obtained in each scenario and the presented results in the first case shows the practical flexibility. Two price adjustment scenarios are defined as follows.

Scenario 1: The price of natural gas is reduced to 1.2 ¢/kWh during electricity peak hours.

Scenario 2: The price of electricity is reduced to 10.75 ¢/kWh during off-peak hours.

The results of the practical flexibility proposals through the incentive-based approach of these two scenarios are shown in Figure 7. According to the defined scenarios, incentive-based price adjustment programs have been able to provide the desired level of practical flexibility for the LDC. This demonstrates the ability of end-use prosumers to provide the flexibility required by the distribution network if appropriate incentive-based DR programs are used.

The main superiorities of the proposed decentralized coordination framework over other frameworks published in the literature should also be demonstrated. It should be noted that three different coordination framework structures are referred to in this discussion. In order to avoid explaining these structures multiple times, they are named and referred to as the first approach, the second approach, the third approach. Their descriptions are:

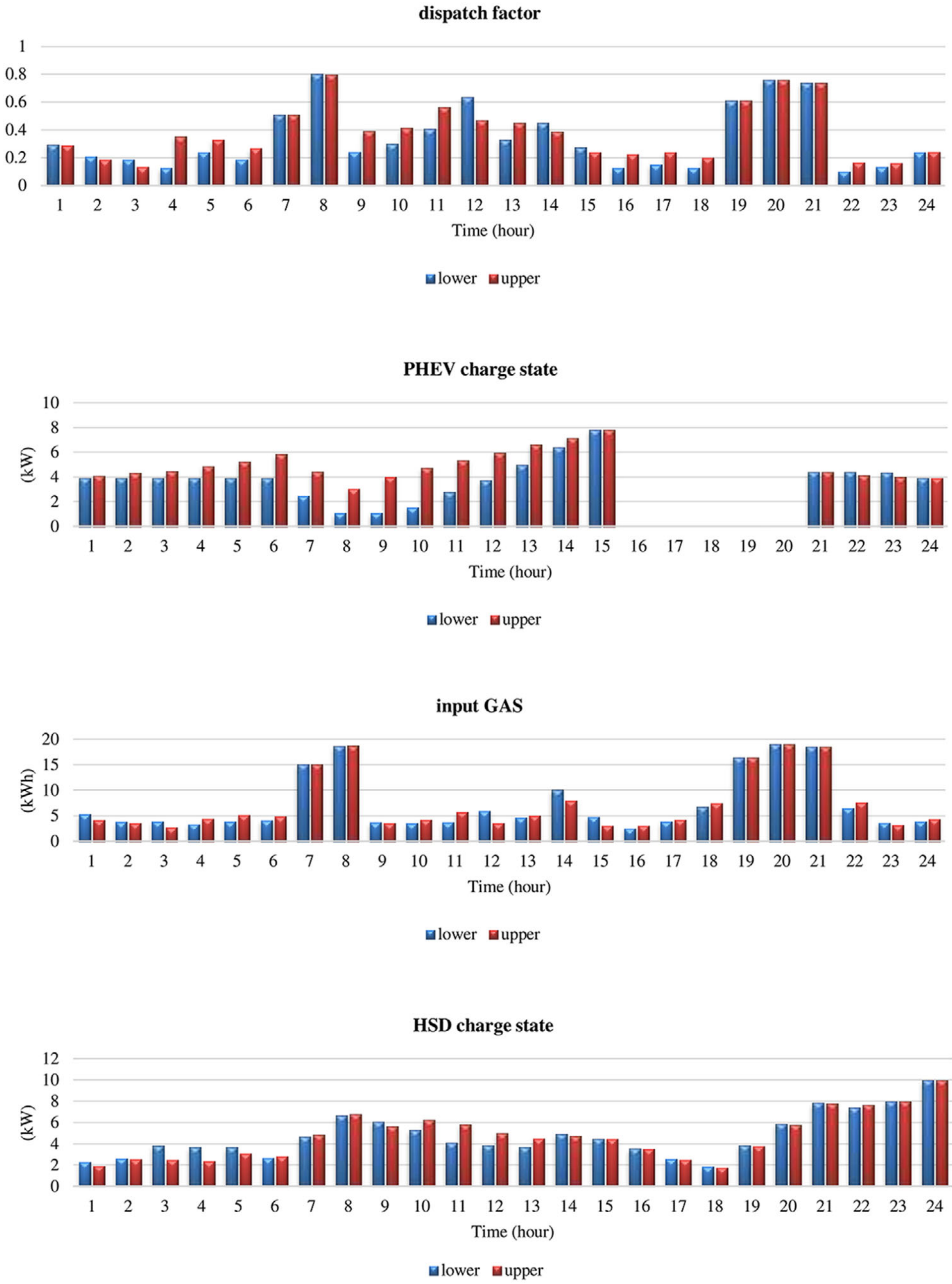

FIGURE 5 The result of the base case

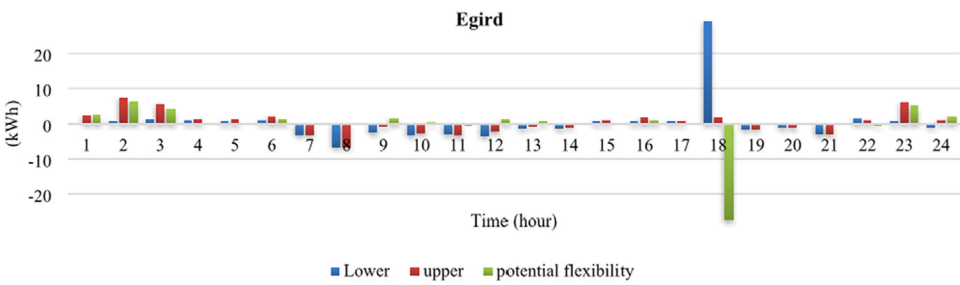

FIGURE 6 The base case load profile and potential flexibility

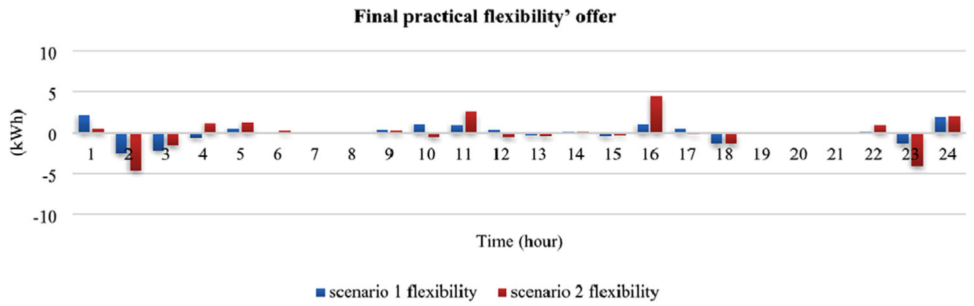

FIGURE 7 The price adjustment programs' practical flexibility

TABLE 6 Total costs of the first and second approaches

Approaches	Total daily costs (¢)
The first approach	806.934
The second approach	720.295

The first approach: The proposed decentralized coordination framework of this paper.

The second approach: Considering a centralized coordination framework, similar to Ref. [50].

The third approach: Considering a decentralized coordination framework, in which the HEMSs sequentially update their load profiles such that system energy cost is minimized, similar to Ref. [19].

At first, as shown in Table 6, it appears that using the proposed framework has resulted in a 12% increase in total cost when compared to the second approach. However, by preserving the privacies of prosumers in the first approach, the prosumers' participation in DR programs is increased. Therefore, their desired load profiles have become very close to the upper-level's announced load profile. As a result, the coverage of the upper level's announced load profile, has been significantly increased in the first approach compared to the second approach, and the improvement of the operation of the grid is become enhanced. Furthermore, as opposed to the third approach, the obtained results are guaranteed to be the global optimum results because the optimization problem is converted to a linear model in the first approach.

4 | CONCLUSION

This paper proposed a new bi-level probabilistic optimization framework between a residential prosumer and an LDC to improve the grid's operation as well as the prosumer's social welfare. Using the proposed approach, residential prosumers' HEMSs optimize the REHs load profiles according to end-users' objective, preferences, and pre-defined parameters. Moreover, the evaluation of the prosumer's potential flexibility is performed through the proposed framework to provide flexibility services, and then prosumers' total practical flexibility' offers are communicated to the LDC, which evaluates operations of the system. These proposals are achieved through the DR incentive-based programs and create no extra cost for the LDC. To compared to centralized approaches that require a lot of simultaneous optimizations in operations, one of the major advantages of the proposed approach is the distribution of computational burden between different entities.

This paper shows that the residential prosumers potentially have the ability to contribute to providing flexibility services of distribution systems. From the results of the implementation of the proposed framework, it is found that there are significant improvements in the total distribution system load profile while maintaining the desired social welfare level of the prosumer. Furthermore, the results revealed that the appropriate DR programs can effectively help DSOs to enhance the operation of distribution systems. Therefore, the proposed approach has the potential to be used for the system operators to manage the load profiles of all available residential prosumers in an efficient way.

FUNDING INFORMATION

None

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

Data sharing not applicable—no new data generated.

ORCID

Moein Moeini-Aghtaie https://orcid.org/0000-0002-7656-6807

REFERENCES

1. Hadi, M.B., Moeini-Aghtaie, M.: Assessing operational flexibility of microgrids considering electrical and thermal energy resources. In: 2019 Smart Grid Conference (SGC). Tehran, Iran, pp. 1–6 (2019)
2. Adefarati, T., Bansal, R.C.: Integration of renewable distributed generators into the distribution system: A review. IET Renewable Power Gener. 10(7), 873–884 (2016)

3. Davoudi, M., Moeini-Aghtaie, M., Ghorani, R.: Developing a new framework for transactive peer-to-peer thermal energy market. IET Gener. Transm. Distrib. 15, 1984–1995 (2021)
4. Ma, J., Silva, V., Belhomme, R., Kirschen, D.S., Ochoa, L.F.: Evaluating and planning flexibility in sustainable power systems. IEEE Trans. Sustainable Energy 4(1), 200–209 (2013)
5. Vegunta, S.C., Watts, C.F.A., Djokic, S.Z., Milanović, J.V.: Review of GB electricity distribution system's electricity security of supply, reliability and power quality in meeting UK industrial strategy requirements. IET Gener. Transm. Distrib. 13(16), 3513–3523 (2019)
6. Davoudi, M., Moeini-Aghtaie, M.: Local energy markets design for integrated distribution energy systems based on the concept of transactive peer-to-peer market. IET Gener. Transm. Distrib. 16, 41–56 (2022)
7. Dadkhah, A., Vahidi, B., Shafie-Khah, M., Catalão, J.P.S.: Power system flexibility improvement with a focus on demand response and wind power variability. IET Renewable Power Gener. 14(6), 1095–1103 (2020)
8. Rajabi, F., Salami, A., Azimi, M.: Hierarchical and multi-level demand response programme considering the flexibility of loads. IET Gener. Transm. Distrib. 14(6), 1051–1061 (2020)
9. Kuzlu, M.: Score-based intelligent home energy management (HEM) algorithm for demand response applications and impact of HEM operation on customer comfort. IET Gener. Transm. Distrib. 9(7), 627–635 (2015)
10. Lopes, D.F., Simões, M., Sampaio, G., Rua, D., Machado, P., Bessa, R.: From home energy management system local flexibility to low voltage predictive grid. In: CIRED 2020 Berlin Workshop (CIRED 2020). pp. 94–97 (2020)
11. Nguyen, H.K., Song, J.B., Han, Z.: Demand side management to reduce peak-to average ratio using game theory in smart grid. In: Proceedings of IEEE Conference on Computing and Communication Workshops (INFOCOM WKSHPS). Orlando, FL, pp. 91–96 (2012)
12. Celik, B., Roche, R., Bouquain, D.: Increasing renewable local energy use in smart neighborhoods through coordinated trading. In: Cyber-Physical-Social Systems and Constructs in Electric Power Engineering. 1st ed. Ch. 9, IET, London (2016)
13. Zhong, H., Xie, L., Xia, Q.: Coupon incentive-based demand response: Theory and case study. IEEE Trans. Power Syst. 28(2), 1266–1276 (2013)
14. Parvania, M., Fotuhi-Firuzabad, M., Shahidehpour, M.: ISO's optimal strategies for scheduling the hourly demand response in day-ahead markets. IEEE Trans. Power. Syst. 29(6), 2636–2645 (2014)
15. Vardakas, J.S., Zorba, N.: A survey on demand response programs in smart grids: Pricing methods and optimization algorithms. IEEE Commun. Surv. Tutor. 17, 152–178 (2015)
16. Celik, B., Roche, R., Bouquain, D., Miraoui, A.: Decentralized Neighborhood Energy Management With Coordinated Smart Home Energy Sharing. IEEE Trans. Smart Grid 9(6), 6387–6397 (2018)
17. Mhana, S., Chapman, A.C., Verbic, G.: A fast distributed algorithm for large-scale demand response aggregation. IEEE Trans. Smart Grid 7(4), 2094–2107 (2016)
18. Mohsenian-Rad, A.H., Wong, V.W.S., Jatskevich, J., Schober, R., Leon-Garcia, A.: Autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid. IEEE Trans. Smart Grid 1,(3), 320–331 (2010)
19. Safdarian, A., Fotuhi-Firuzabad, M., Lehtonen, M.: A distributed algorithm for managing residential demand response in smart grids. IEEE Trans. Ind. Inf. 10(4), 2385–2393 (2014)
20. Alrumayh, O., Bhattacharya, K.: Flexibility of residential loads for demand response provisions in smart grid. IEEE Trans. Smart Grid 10(6), 6284–6297 (2019)
21. Samadi, P., Mohsenian-Rad, H., Schober, R., Wong, V.W.S.: Advanced demand side management for the future smart grid using mechanism design. IEEE Trans. Smart Grid 3(3), 1170–1180 (2012)
22. Li, N., Chen, L., Low, S.H.: Optimal demand response based on utility maximization in power networks. In: IEEE Power Energy Society General Meeting. Detroit, MI, pp. 1–8 (2011)
23. Gatsis, N., Giannakis, G.B.: Residential load control: Distributed scheduling and convergence with lost AMI messages. IEEE Trans. Smart Grid 3(2), 770–786 (2012)
24. Rastegar, M., Fotuhi-firuzabad, M.: Outage management in residential demand response programs. IEEE Trans. Smart Grid 6,(3), 1453–1462 (2015)
25. Erdinc, O., Paterakis, N.G., Mendes, T.D.P., Bakirtzis, A.G., Catalão, J.P.S.: Smart household operation considering bi-directional EV and ESS utilization by real-time pricing-based DR. IEEE Trans. Smart Grid 6(3), 1281–1291 (2015)
26. Fotuhi-Firuzabad, M., Rastegar, M., Safdarian, A., Aminifar, F.: Probabilistic home load controlling considering plug-in hybrid electric vehicle uncertainties. In: Reliability Modeling and Analysis of Smart Power Systems. pp. 117–132. Springer, New York (2014)
27. Chen, X., Wei, T., Hu, S.: Uncertainty-aware household appliance scheduling considering dynamic electricity pricing in smart home. IEEE Trans. Smart Grid 4(2), 932–941 (2013)
28. Liu, M., Lee, W.-J., Lee, L.K.: Financial opportunities by implementing renewable sources and storage devices for households under ERCOT demand response programs design. IEEE Trans. Ind. Appl. 50(4), 2780–2787 (2014)
29. Brahman, F., Honarmand, M., Jadid, S.: Optimal electrical and thermal energy management of a residential energy hub, integrating demand response and energy storage system. Energy Build. 90, 65–75 (2014)
30. Zheng, X., Chen, H., Xu, Y., Liang, Z., Chen, Y.: A Hierarchical Method for Robust SCUC of Multi-Area Power Systems with Novel Uncertainty Sets. IEEE Trans. Power Syst. 35(2), 1364–1375 (2020)
31. Taheri, S., Ghoraani, R., Pasban, A., Moeini-Aghtaie, M., Safdarian, A.: Stochastic framework for planning studies of energy systems: A case of EHs. IET Renewable Power Gener. 14(3), 435–444 (2019)
32. Zheng, X., Qu, K., Lv, J., Li, Z., Zeng, B.: Addressing the conditional and correlated wind power forecast errors in unit commitment by distributionally robust optimization. IEEE Trans. Sustainable Energy 12(2), 944–954 (2021)
33. Xu, Z., et al.: Performance analysis and comparison on energy storage devices for smart building energy management. IEEE Trans. Smart Grid 3(4), 2136–2147 (2012)
34. Verbic, G., Cañizares, C.A.: Probabilistic optimal power flow in electricity markets based on a two-point estimate method. IEEE Trans. Power Syst. 21(4), 1883–1893 (2006)
35. Hahn, G.J., Shapiro, S.S.: Statistical Models in Engineering. Wiley-Interscience, Hoboken, NJ (1994)
36. Kaplan, S.: On the method of discrete probability distributions in risk and reliability calculation-application to seismic risk assessment. Risk Anal. 1(3), 189–196 (1981)
37. Zhang, Q.: A general method dealing with correlations in uncertainty propagation in fault trees. Reliab. Eng. Syst. Saf. 26(3), 231–247 (1989)
38. Schellenberg, A., Rosehart, W., Aguado, J.: Cumulant-based probabilistic optimal power flow (P-OPF) with Gaussian and gamma distributions. IEEE Trans. Power Syst. 20(2), 773–781 (2005)
39. Tian, W.D., Sutanto, D., Lee, Y.B., Outhred, H.R.: Cumulant based probabilistic power system simulation using Laguerre polynomials. IEEE Trans. Energy Convers. 4(4), 567–574 (1989)
40. Malekpour, A.R., Niknam, T.: A probabilistic multi-objective daily Volt/Var control at distribution networks including renewable energy sources. J. Energy 36(5), 3477–3488 (2011)
41. Aien, M., Fotuhi-Firuzabad, M., Rashidinejad, M.: Probabilistic optimal power flow in correlated hybrid wind–photovoltaic power systems. IEEE Trans. Smart Grid 5(1), 130–138 (2014)
42. Li, N., Chen, L., Low, S.H.: Optimal demand response based on utility maximization in power networks. In: Proceedings of IEEE Power Energy Society General Meeting. San Diego, CA, pp. 1–8 (2011)
43. Su, C.-L.: Probabilistic load-flow computation using point estimate method. IEEE Trans. Power Syst. 20(4), 1843–1851 (2005)
44. Kovačević, R.P.B.: Federal trade commission. Phil. Mag. 27, 77 (2009)
45. Website: End-use consumption of electricity 2001. Phil. Mag. 27, 77 (2009)
46. Shafiee, S., Fotuhi-Firuzabad, M., Rastegar, M.: Investigating the impacts of plug-in hybrid electric vehicles on distribution congestion. In: 22nd International Conference and Exhibition on Electricity Distribution (CIRED 2013). Stockholm, Sweden, pp. 1–4 (2013)

47. Linna, N., Changsen, F., Wen, F., Salam, A.: Optimal power flow of multiple energy carriers with multiple kinds of energy storage. In: IEEE Power Energy Society General Meeting. Boston, MA, pp. 145–155 (2016)
48. Erdinc, O.: Economic impacts of small-scale own generating and storage units, and electric vehicles under different demand response strategies for smart households. Appl. Energy 126, 142–150 (2014)
49. Gelaro et al.: Ground level solar irradiance. renewables.ninja, (2019). Accessed on: 20th January, 2021. http://www.renewables.ninja
50. Oliver Kriett, P., Salani, M.: Optimal control of a residential microgrid. Energy 42(1), 321–330 (2012)

How to cite this article: Ostovar, S., Moeini-Aghtaie, M., Hadi, M.B.: Flexibility provision of residential energy hubs with demand response applications. IET Gener. Transm. Distrib. 16, 1668–1679 (2022). https://doi.org/10.1049/gtd2.12392